

EXPERIMENT 01: ER DIAGRAM

AIM

To design an Entity-Relationship (ER) diagram for an Indian e-commerce platform consisting of Customer, Product, Order Item, Seller, Category, Payment, Delivery, and Address entities. To identify primary keys, composite attributes, multivalued attributes, weak entities, specialization of Product, cardinality ratios, and participation constraints.

THEORY

An Entity-Relationship (ER) diagram is a graphical representation of entities, their attributes, and the relationships among them in a database system. It is used during database design to represent the structure of a database clearly and systematically.

For an Indian e-commerce platform, the ER diagram represents customers purchasing products from sellers, making payments, and receiving deliveries at their addresses.

ER DIAGRAM COMPONENTS

S. No.	Component	Representation
1	Strong Entity	Single Rectangle
2	Weak Entity	Double Rectangle
3	Attribute	Oval
4	Multivalued Attribute	Double Oval
5	Composite Attribute	Oval connected to sub-attributes
6	Derived Attribute	Dashed Oval
7	Relationship	Diamond
8	Identifying Relationship	Double Diamond
9	Partial Participation	Single Line
10	Total Participation	Double Line
11	Cardinality	1:1, 1:N, M:N

ENTITIES AND ATTRIBUTES

The ER diagram consists of the following entities:

1. Customer

- Primary Key: Customer_ID
- Attributes: Name, Email, Phone_Number, Date_of_Birth
- Composite Attribute: Name (First_Name, Middle_Name, Last_Name)
- Multivalued Attribute: Phone_Number

2. Product

- Primary Key: Product_ID
- Attributes: Product_Name, Price, Description, Stock_Quantity

- Specialization: Electronics, Clothing, and Grocery

3. Order Item (Weak Entity)

- Partial Key: Item_Number
- Attributes: Quantity, Unit_Price, Subtotal
- Depends on the associated Order for identification.
- An Order Item can be identified using Order_ID and Item_Number together.

4. Seller

- Primary Key: Seller_ID
- Attributes: Seller_Name, Email, Phone_Number, Business_Name

5. Category

- Primary Key: Category_ID
- Attributes: Category_Name, Description

6. Payment

- Primary Key: Payment_ID
- Attributes: Payment_Date, Amount, Payment_Method, Payment_Status
- Payment methods may include UPI, credit card, debit card, net banking, and cash on delivery.

7. Delivery

- Primary Key: Delivery_ID
- Attributes: Tracking_Number, Delivery_Status, Dispatch_Date, Delivery_Date

8. Address

- Primary Key: Address_ID
- Attributes: House_Number, Street, City, State, PIN_Code, Country
- Composite Attribute: Address, consisting of House_Number, Street, City, State, PIN_Code, and Country.

RELATIONSHIPS AND CARDINALITY

S. No.	Relationship	Cardinality	Description
1	Customer – Order	1:N	A customer can place multiple orders.
2	Order – Order Item	1:N	An order can contain multiple order items.
3	Product – Order Item	1:N	A product can appear in multiple order items.
4	Seller – Product	1:N	A seller can sell multiple products.
5	Category – Product	1:N	A category can contain multiple products.
6	Order – Payment	1:N	An order may have multiple payment records or attempts.
7	Order – Delivery	1:N	An order may involve multiple delivery records or shipments.

S. No.	Relationship	Cardinality	Description
8	Customer – Address	1:N	A customer can have multiple addresses.
9	Order – Address	N:1	Multiple orders can use the same delivery address.

KEY CONCEPTS USED

- 1. Primary Key:** A primary key uniquely identifies each record in an entity. Examples include Customer_ID, Product_ID, Seller_ID, and Payment_ID.
- 2. Composite Attribute:** An attribute that can be divided into smaller sub-attributes. For example, Address can be divided into House_Number, Street, City, State, PIN_Code, and Country.
- 3. Multivalued Attribute:** An attribute that can have multiple values for one entity. For example, a customer may have multiple phone numbers.
- 4. Weak Entity:** An entity that depends on another entity for identification. Order Item is a weak entity because it depends on Order for its identification.
- 5. Specialization:** A process of dividing a general entity into more specific sub-entities. Product can be specialized into Electronics, Clothing, and Grocery.
- 6. Cardinality:** Cardinality specifies the number of entities that can participate in a relationship, such as 1:1, 1:N, and M:N.
- 7. Participation Constraints:** Participation constraints specify whether the participation of an entity in a relationship is total or partial. Total participation is represented by a double line, whereas partial participation is represented by a single line.

ER DIAGRAM

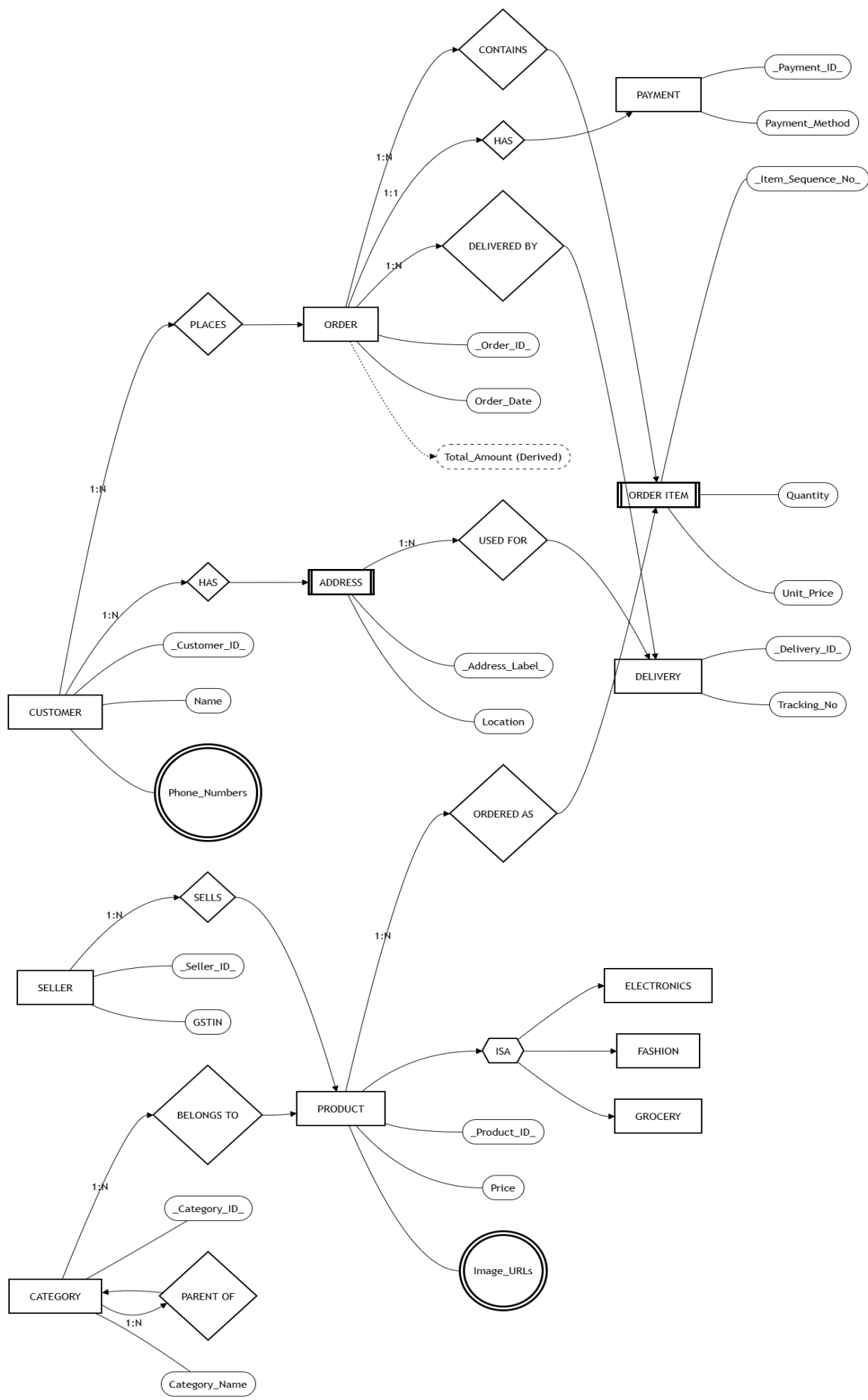


Figure 1: ER Diagram of an Indian E-commerce Platform

RESULT

The ER diagram for an Indian e-commerce platform was designed using standard ER-model symbols. It represents the entities, attributes, primary keys, relationships, weak entities, composite and multivalued attributes, Product specialization, cardinality ratios, and participation constraints required for database design.